

Section 1: Counting

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1 Introduction

1.1 Logistics

Welcome! The goal of our weekly section is to review content from lecture, with emphasis on big-picture intuition, mathematical derivation, and hands-on practice.

Specifically, we aim for section to be interactive, well-paced, inclusive, and fun! Please do not hesitate to reach out if you have any questions/feedback!

1.2 Office Hours

- See the Google Calendar for the most accurate information!

1.3 Attendance

- bit.ly/stat110harvard

2 Set Theory

Idea. *Sets* are the foundation of modern mathematics, including probability theory. Much of the language of this course is in set theory.

Definition 1 (Set). A *set* S is a collection of elements.

- $x \in S$ means x is an element of the set S (while $x \notin S$ means x is *not* in S).
- The *cardinality* of S , denoted $|S|$, is its size (i.e., the number of unique elements).
- E.g., if $S = \{1, 3, 5, \dots\}$ is the set of all odd numbers, then $1 \in S$ while $2 \notin S$.

Definition 2 (Empty set). The *empty set*, denoted $\emptyset$ or $\{\}$, is the set with no elements.

Definition 3 (Subset). If A and B are sets, then A is a *subset* of B , denoted $A \subseteq B$, if every element of A is also an element of B .

- E.g., if $S = \{1, 3, 5, \dots\}$ is the set of all odd numbers, then $S \subseteq \mathbb{N}$, where $\mathbb{N} = \{1, 2, 3, \dots\}$ is the set of natural numbers.

Definition 4 (Union). The *union* of two sets A and B , denoted $A \cup B$, is the set of all elements in A or B (or both):

$$A \cup B = \{x : x \in A \text{ or } x \in B\}.$$

Definition 5 (Intersection). The *intersection* of two sets A and B , denoted $A \cap B$, is the set of all elements that are in both A and B :

$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$

- We say A and B are *disjoint* if $A \cap B = \emptyset$.

Definition 6 (Complement). Let $A \subseteq S$. The *complement* of A , denoted A^c , is the set of all elements in S that are not in A :

$$A^c = \{x \in S : x \notin A\}.$$

Definition 7 (De Morgan's Laws). We can “distribute” the *complement* inside parentheses if we “flip” the operation:

$$(A_1 \cup A_2 \cup \dots \cup A_n)^c = A_1^c \cap A_2^c \cap \dots \cap A_n^c$$

and

$$(A_1 \cap A_2 \cap \dots \cap A_n)^c = A_1^c \cup A_2^c \cup \dots \cup A_n^c.$$

- E.g., $(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$.

 **Concept Checker 1.** Let $S = \{1, 3, 5, \dots\}$ be the set of all odd numbers and $\mathbb{N} = \{1, 2, 3, \dots\}$ be the set of natural numbers. What is $S \cup \mathbb{N}$? What is $S \cap \mathbb{N}$? For $S \subseteq \mathbb{N}$, what is S^c ?

Solution

$S \cup \mathbb{N} = \mathbb{N} = \{1, 2, 3, \dots\}$, $S \cap \mathbb{N} = S = \{1, 3, 5, \dots\}$, and $S^c = \{2, 4, 6, \dots\}$ (i.e., the set of all even numbers excluding 0).

3 Probability

Idea. We'll introduce important definitions so that we can rigorously discuss *probability*, beginning with the naive definition. As we'll see, naive probability is limited in its assumptions.

Definition 8 (Outcome). An *outcome* is the result after conducting an experiment.

- E.g., I get dealt the ace of hearts.

Definition 9 (Sample space). The *sample space* S of an experiment is the set of all possible *outcomes*.

- E.g., the set of the 52 cards I could've been dealt.

Definition 10 (Event). An *event* A is a collection of *outcomes*.

- Formally, an event is a *subset* of the *sample space* (i.e., $A \subseteq S$).
- We say that A *occurred* if the actual/observed *outcome* is in A .
- E.g., the event that I get dealt an ace is a collection of 4 outcomes.

Definition 11 (Naive definition of probability). Let A be an *event* for an experiment with a finite *sample space* S . The *naive probability* of A is

$$P_{\text{naive}}(A) = \frac{\# \text{ of outcomes favorable to } A}{\# \text{ of total outcomes in } S} = \frac{|A|}{|S|}.$$

- $\otimes$: We can only use this when the sample space is *finite* and consists of *equally likely outcomes*.
- E.g., let A be the event that I get dealt an ace. Then $P(A) = \frac{4}{52} = \frac{1}{13}$.

✎ Concept Checker 2. Why is it okay to use the naive definition of probability in the example above (where A is the event I get dealt an ace)? What is an example where it is not okay?

Solution

Here, the sample space is finite and consists of equally likely outcomes. Consider the probability of life on Mars. Since $S = \{\text{life, no life}\}$, $P_{\text{naive}}(\text{life}) = \frac{1}{2}$, but this is nonsensical since these should not be viewed as equally likely outcomes!

4 Counting

Idea. Naive probability requires *counting* the number of possibilities, so we must discuss combinatorics, the branch of mathematics concerning counting. The foundation of all the formulas and techniques we'll introduce is the *multiplication rule*.

4.1 Counting Techniques

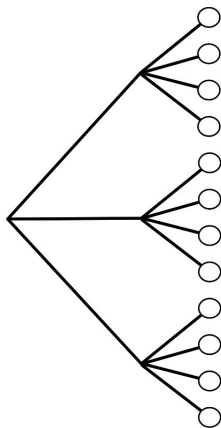


FIGURE 1: A tree diagram illustrating the multiplication rule.

Definition 12 (Multiplication rule). Suppose that Experiment A has a possible outcomes, and for each of those outcomes, Experiment B has b possible outcomes. Then the *compound experiment* has $a \times b$ possible outcomes.

- In Figure 1, the tree diagram illustrates an example where Experiment A has 3 possible outcomes, for each of which Experiment B has 4 possible outcomes, leading to $3 \times 4 = 12$ possible outcomes overall.

Definition 13 (Factorial). For any positive integer n ,

$$n! = n(n-1)(n-2) \cdots 1.$$

- For $n = 0$, $0! = 1$.

Definition 14 (Permutation). For $1, 2, \dots, n$, there are $n!$ *permutations*, each of which is an arrangement in some order.

- E.g., for $(1, 2, 3, 4, 5)$, there are $5! = 120$ permutations, of which $(3, 5, 1, 2, 4)$ is one.

 **Concept Checker 3.** Suppose that there is a race with 3 people: Alice, Bob, and Charlie. How many possibilities are there for the first, second, and third place winners?

Solution

Using the definition of permutation, there are $3! = 6$ arrangements. Using the multiplication rule, there are 3 possibilities for who gets first place. Once that is fixed, there are 2 possibilities for who gets second place. Once those are fixed, there is 1 possibility for who gets third place. Thus, $3 \times 2 \times 1 = 6$.

Definition 15 (Adjusting for overcounting). It is not always easy to directly count each possibility only once. But if we are able to count each possibility exactly c times (for some c), then we can adjust by *dividing by c* .

-  Ask whether or not *order matters* since, if it doesn't, we often need to *adjust for overcounting*.
- E.g., in *rankings*, order matters, so there are $3 \times 2 = 6$ ways to choose first and second place in a 3-person race. But in *groupings*, order doesn't matter, so we double-counted each possibility and divide by 2 to get the correct count (i.e., $\frac{6}{2} = 3$ ways to choose groups of 2).

Definition 16 (Binomial coefficient). For any nonnegative integers k and n , the *binomial coefficient* $\binom{n}{k}$ is the number of subsets of size k for a set of size n .¹

- For $k \leq n$,

$$\binom{n}{k} = \frac{n(n-1) \cdots (n-k+1)}{k!} = \frac{n!}{(n-k)!k!}.$$

- For $k > n$, $\binom{n}{k} = 0$.
- Sets and subsets are *unordered* (e.g., $\{1, 2\} = \{2, 1\}$), so we are counting the *number of ways to choose k objects out of n , without replacement and where order doesn't matter*. Thus, we divide by $k!$ to *adjust for overcounting*.
- E.g., there are $\binom{3}{2} = 3$ ways to choose 2 people from a group of 3.

 **Concept Checker 4.** Consider a group of 4 people. How many ways are there to choose a 2-person committee? How many ways are there to break everyone into 2 teams of 2?

Solution

First, using the definition of binomial coefficient, there are $\binom{4}{2} = 6$ ways to choose 2 people from a group of 4. Using the multiplication rule, there are 4 ways to choose the first person and 3 ways to choose the second person, but we divide by 2 to adjust for

¹This is sometimes referred to as a *combination*.

overcounting (since, e.g., the committee $\{A, B\}$ is the same as the committee $\{B, A\}$), leading to the same answer of $\frac{4 \times 3}{2} = 6$. Second, using the previous result, there are $\frac{6}{2} = 3$ ways to break everyone into 2 teams of 2. We divide by 2 to adjust for overcounting (since, e.g., choosing the team $\{A, B\}$ implies the other team is $\{C, D\}$).

 **Concept Checker 5.** Suppose that you can go to either Jefe's or Felipe's, and at each place, you can order a burrito, quesadilla, or nachos. How many possible meal choices are there? If you eat once in the morning and again in the afternoon, how many possibilities are there now? Finally, if you only care about what foods you can get in the day (and not the order in which you get them), how many possibilities are there now?

Solution

Let $R_t(o)$ correspond to order o at restaurant R at time t (e.g., $F_m(b)$ means Felipe's burrito in the morning). First, there are 2 choices for restaurants and, at each, 3 choices for orders, so there are $2 \times 3 = 6$ possibilities. Next, there are now 2 choices for time, so at each previous possibility (e.g., $J(b)$), there are 6 more branches that correspond to the afternoon options (e.g., after $J_m(b)$ can be $J_a(b), J_a(q), \dots$). Thus, there are $2 \times 3 \times 6 = 36$ possibilities. Finally, if we only care about what foods we can get in the day, then we're looking for unordered possibilities. E.g., $J_m(b), J_a(q)$ and $J_m(q), J_a(b)$ should be counted only once as $J(b), J(q)$. We cannot simply divide 36 by 2 since some possibilities repeat and are only listed once (e.g., $J_m(b), J_a(b)$). There are 6 of these repeaters, so the number of ordered possibilities that aren't the same is $36 - 6 = 30$. We double count these, so the number of unordered possibilities that aren't the same is $\frac{30}{2} = 15$. We add the 6 repeaters that are only listed once, so there are $15 + 6 = 21$ possibilities.

4.2 Sampling

Let $1 \leq k \leq n$. How many possible samples of size k are there from n objects?

	Order Matters	Order Doesn't Matter
W/ Replacement	$n^k = \underbrace{n \times n \times \dots \times n}_k$	$\binom{n+k-1}{k-1}$
W/o Replacement	$\frac{n!}{(n-k)!} = \underbrace{n(n-1) \dots (n-k+1)}_k$	$\binom{n}{k} = \frac{n(n-1) \dots (n-k+1)}{k!} = \frac{n!}{(n-k)!k!}$

Example 1 (Order matters, w/ replacement). We have a jar with n balls, numbered 1 to n . Each time we take a ball out, we return it to the jar. Thus, there are n possibilities for each draw, so there are n^k ways to obtain a sample of size k .

- E.g., there are $3^2 = 3 \times 3 = 9$ ways to choose who is first in one race and who is first in another race ($k = 2$), each with 3 people ($n = 3$).

Example 2 (Order matters, w/o replacement). We have a jar with n balls, numbered 1 to n . Each time we take a ball out, we *don't* return it to the jar. Thus, there are n possibilities for the first sampled ball, $n - 1$ possibilities for the second sampled ball, etc., so there are $n(n - 1) \cdots (n - k + 1)$ ways to obtain a sample of size k .

- E.g., there are $\frac{3!}{(3-2)!} = 3 \times 2 = 6$ ways to choose who is first and who is second ($k = 2$) in a race with 3 people ($n = 3$).

Example 3 (Order doesn't matter, w/ replacement). This counting method is *not* covered in STAT 110 this year!

Example 4 (Order doesn't matter, w/o replacement). We have a jar with n balls, numbered 1 to n . Each time we take a ball out, we *don't* return it to the jar. Thus, there are n possibilities for the first sampled ball, $n - 1$ possibilities for the second sampled ball, etc., but if we *don't* distinguish between order (e.g., $\{1, 2\} = \{2, 1\}$), then we divide by $k!$ to avoid overcounting different permutations of the same group, so there are $\frac{n(n-1)\cdots(n-k+1)}{k!}$ ways to obtain a sample of size k .

- E.g., there are $\binom{3}{2} = \frac{3 \times 2}{2!} = \frac{3!}{(3-2)!2!} = 3$ ways to choose which pairs get top 2 ($k = 2$) in a race with 3 people ($n = 3$).

 **Concept Checker 6.** What is the difference between the “Order Matters, w/o Replacement” and “Order Doesn't Matter, w/o Replacement”—mathematically and intuitively?

Solution

Mathematically, “Order Doesn't Matter, w/o Replacement”—i.e., $\binom{n}{k}$ —takes “Order Matters, w/o Replacement”—i.e., $\frac{n!}{(n-k)!}$ —and divides by $k!$. Intuitively, if order doesn't matter, we're overcounting by the $k!$ permutations of the same group. Consider choosing 2 people out of 3 (i.e., $n = 3$ and $k = 2$). There are $3 \times 2 = 6$ ways to choose who is first and who is second (i.e., order matters such that $\{\text{Alice}, \text{Bob}\} \neq \{\text{Bob}, \text{Alice}\}$), but if we only care about choosing which pairs get top two, then we've overcounted by $2! = 2$ (i.e., order doesn't matter such that $\{\text{Alice}, \text{Bob}\} = \{\text{Bob}, \text{Alice}\}$). Thus, there are $\binom{3}{2} = \frac{3 \times 2}{2} = 3$ ways to choose which pairs get top 2.

 **Concept Checker 7.** How many ways are there to permute the letters in “GUSH”? How about in “GUUUSH”?

Solution

First, there are $4! = 24$ permutations of “GUSH”. Using the table, we are sampling $k = 4$ from $n = 4$ without replacement and where order matters. Using the multiplication rule, we can illustrate this with a tree diagram. Second, there are $\binom{6}{3} \binom{3}{1} \binom{2}{1} \binom{1}{1} = \frac{6!}{3!} = 120$ permutations of “GUUUSH”. One method is to choose where to put the 3 U's, then where to put the 1 G (from the remaining spots), then where to put the 1 S, and then where to put the 1 H. Another method is to start with $6!$ and adjust for overcounting, dividing by $3!$ since the 3 U's can be permuted among themselves in any

way (e.g., “GU₁U₂U₃SH” is effectively the same as “GU₃U₂U₁SH”).

5 Story Proofs

Idea. Formal mathematical proofs can be difficult, so often, it is easier (and better for building intuition) to use *story proofs*, which will appear throughout the course.

Definition 17 (Story proof). A proof by interpretation or application rather than algebra or calculus.

Example 5 (Symmetry rule for binomial coefficients). $\binom{n}{k} = \binom{n}{n-k}$.

Proof. If we want to choose k people out of n , we can either directly pick the k people to be included (left-hand side) or pick the $n - k$ people to be not included (right-hand side). E.g., there are $\binom{3}{2} = 3$ ways to choose which pairs get top 2 ($k = 2$) in a race with 3 people ($n = 3$). Equivalently, there are $\binom{3}{1} = 3$ ways to choose which person does not get top 2 ($n - k = 1$). $\square$

 **Concept Checker 8.** Show that for $k > n$, $\binom{n}{k} = 0$ via story proof.

Solution

E.g., let $k = 3$ and $n = 2$. How many ways are there to choose a group of 3 from 2 people? None!

 **Concept Checker 9.** Show that $\binom{r}{m}\binom{m}{k} = \binom{r}{k}\binom{r-k}{m-k}$ via story proof.

Solution

On the left-hand side, we pick a team (of size m) from a class (of size r) and then pick k leaders from the team. On the right-hand side, we pick the k leaders directly from the class (of size r) and then, from the remaining classmates (of which there are $r - k$), pick the non-leader teammates (of which there are $m - k$).

6 Exercises

 **Problem 1.** *Connections* is a popular puzzle published by *The New York Times*, where players are presented with a grid of 16 words and must organize them into 4 distinct groups (each of 4 words) that share a hidden common theme. These categories are color-coded by difficulty, with yellow being the easiest and purple being the hardest. The player wins once they successfully find 3 connections (since the last one is deterministic), but they are only allowed 4 mistakes across the entire puzzle.

(a) If a person randomly guesses 4 words at the beginning, what is the probability that they will find a connection?

- (b) If a person randomly guesses 4 words at the beginning, what is the probability that they will find the purple connection?
- (c) Your friend claims that they found all 4 connections with no mistakes by randomly guessing. How likely is this?

Solution

- (a) Using the naive definition of probability, $p = \frac{4}{\binom{16}{4}} = \frac{1}{455} = 0.00219$. There are $\binom{16}{4}$ ways to choose an unordered group of 4 from the 16 words, and 4 of those are connections.
- (b) Using the naive definition of probability, $p = \frac{1}{\binom{16}{4}} = \frac{1}{1820} = 0.00054$. There are $\binom{16}{4}$ ways to choose an unordered group of 4 from the 16 words, and 1 of those is the purple connection.
- (c) Using the naive definition of probability, $p = \frac{1}{\binom{16}{4}\binom{12}{4}\binom{8}{4}/4!} = \frac{1}{2627625}$, which is extremely unlikely. If a player just randomly divides the 16 words into 4 distinct groups (each of 4 words), there are $\binom{16}{4}\binom{12}{4}\binom{8}{4}/4!$ ways to do this, and only 1 of those is the correct partition corresponding to 4 connections. Alternatively, $p = \frac{4}{\binom{16}{4}} \times \frac{3}{\binom{12}{4}} \times \frac{2}{\binom{8}{4}} = \frac{1}{2627625}$. Since the player wins once they find any 3 of the 4 connections, we need the first 3 random guesses to all be correct. For the first guess, there are 4 correct choices out of $\binom{16}{4}$ possibilities. After 1 connection is found, there are 3 correct choices out of $\binom{12}{4}$ possibilities. After 2 connections are found, there are 2 correct choices out of $\binom{8}{4}$ possibilities.

 **Problem 2.** Suppose that an item from Chipotle is constructed by making the following choices:

- type: burrito, burrito bowl, soft taco, crispy taco, salad
- rice: white, brown
- beans: black, pinto
- meat: chicken, steak, carnitas, barbacoa
- vegetable: lettuce, onions
- topping: cheese, guacamole, sour cream
- salsa: corn, green, red, tomato.

One type is chosen, but for the other categories, any combination is possible (e.g., the item can have up to all four types of salsa, or it can have no salsa). For simplicity, assume that ordering a taco gives only one taco (rather than being a set of 3 tacos where different ingredients could be added to each).²

- (a) A hungry person orders an item. How many possibilities are there for what the item is?
- (b) A very hungry person orders 2 items. How many possibilities are there for what the 2 items are?

²Inspired by Problem 1 in “Stat 110 Homework 1, Fall 2024” by Joseph K. Blitzstein.

Solution

(a) There are $5 \times 2^2 \times 2^2 \times 2^4 \times 2^2 \times 2^3 \times 2^4 = 655360$ possibilities. We must choose 1 of the 5 types, but for each of the other categories, there are 2^n possibilities (e.g., for rice, we can have no rice, white rice, brown rice, or white rice and brown rice, with $2^2 = 4$ possibilities).

(b) Notice that this is very similar to the Jefe's/Felipe's problem (i.e., Concept Checker 5). There are 2 choices for order (i.e., first item vs. second item), so at each previous possibility, there are 655360 more branches that correspond to the second item. But we only care about what items we can get, so we're looking for unordered possibilities. E.g., burrito with nothing in it as the first item and burrito bowl with nothing in it as the second item is the same as burrito bowl with nothing in it as the first item and burrito with nothing in it as the second item. We cannot simply divide 655360^2 by 2 since some possibilities repeat and are only listed once (e.g., burrito with nothing in it as the first and second item). There are 655360 of these repeaters, so the number of unordered possibilities that aren't the same is $655360^2 - 655360$. We double count these, so the number of unordered possibilities that aren't the same is $\frac{655360^2 - 655360}{2}$. We add the 655360 repeaters that are only listed once, so there are $\frac{655360^2 - 655360}{2} + 655360 = 214748692480$ possibilities. Alternatively, there are $\binom{655360}{2}$ ways to choose unordered pairs of orders, and we add the 655360 orders that are repeats to get $\binom{655360}{2} + 655360 = 214748692480$ possibilities.

 **Problem 3.** Pineapple Computer hopes to design an amazing new tablet. They split 42 engineers into 3 teams of 14. Each team will create a design independently, and then the company will choose the best of the 3 designs.³

(a) Assume that we don't distinguish between teams (e.g., $\{A, B, \dots, N\}$ being the "first" team is the same as their being the "last" team). How many possibilities are there for the compositions of the teams?

(b) For this part only, assume that instead the 42 engineers are split into a team of 6, a team of 12, and a team of 24 (since the company is interested in whether a larger team will come up with a better design). How many possibilities are there for the compositions of the teams?

(c) 2 of the 42 engineers, Amos and Daniel, are best friends. Suppose that for this part the teams will be kept in separate silos: Silo 1, Silo 2, and Silo 3. The creation of teams and the assignments to silos are completely random (with all possibilities equally likely). Find the probability that Amos and Daniel will work in the same silo, using the naive definition of probability where the denominator counts all possible ways to divide the engineers into silos.

(d) Solve (c) in a simpler way: Imagine that each silo has 14 seats and that we first fix Amos' seat and then randomly choose Daniel's seat.

³Inspired by Problem 2 in "Stat 110 Homework 1, Fall 2024" by Joseph K. Blitzstein.

Solution

(a) There are $\frac{\binom{42}{14}\binom{28}{14}\binom{14}{14}}{3!}$ possibilities. First, there are $\binom{42}{14}$ ways to choose an unordered group of size 14 from the 42 engineers, and once those are fixed, there are $\binom{28}{14}$ ways to choose an unordered group of size 14 from the remaining 28. The final group is deterministic (i.e., $\binom{14}{14} = 1$). But the teams are indistinguishable, so we divide by $3!$ to adjust for overcounting (i.e., $\binom{42}{14}\binom{28}{14}\binom{14}{14}$ constructs the teams sequentially, so each final grouping is counted $3!$ times, once for every ordering of the same three teams).

(b) There are $\binom{42}{6}\binom{36}{12}\binom{24}{24}$ possibilities. Notice that the teams are distinct sizes, so we don't divide by $3!$.

(c) First, the number of possible ways to divide the engineers into silos is $\binom{42}{14}\binom{28}{14}\binom{14}{14}$. Notice that we don't divide by $3!$ since we distinguish between silos (i.e., there are $\binom{42}{14}$ ways to choose those in Silo 1 and, afterwards, $\binom{28}{14}$ ways to choose those in Silo 2). Next, we want to count the number of possibilities where Amos and Daniel work in the same silo. Let's begin with the possibilities where they work in Silo 1. Once that is fixed, there are $\binom{40}{12}$ ways to choose the remaining engineers in Silo 1, and once those are fixed, there are $\binom{28}{14}$ ways to choose those in Silo 2. Silo 3 is deterministic (i.e., $\binom{14}{14} = 1$). Amos and Daniel could've worked in any of the 3 silos, so we multiply by 3. Thus, the probability is $\frac{3\binom{40}{12}\binom{28}{14}\binom{14}{14}}{\binom{42}{14}\binom{28}{14}\binom{14}{14}} = \frac{3\binom{40}{12}}{\binom{42}{14}} = \frac{13}{41}$.

(d) Once Amos is placed in a silo, there are 41 remaining spots (or "seats") Daniel could occupy, and 13 of them are in Amos's silo. Thus, the probability is $\frac{13}{41}$.